# Sumario

- Protocolos MAC: Antecedentes
- Origen de Ethernet
- El nivel de enlace en Ethernet
- Estandarización



Steve Crocker
(1944-

# Ethernet experimental



Robert Metcalfe
(1946-

- En 1972 Robert Metcalfe, ex-estudiante del MIT, entró a trabajar en el Xerox PARC (Palo Alto Research Center), cerca de San Francisco. Le encargaron diseñar la red del laboratorio.
- Leyendo un artículo sobre Alohanet encontró errores en el diseño, lo modificó, lo mejoró y escribió su tesis doctoral
- 22/5/1973: Metcalfe publicó un memorándum en Xerox
- 11/11/1973: Metcalfe y Boggs pusieron en marcha la Ethernet experimental: 2,94 Mbps, direcc. 8 bits, CRC 16 bits, protocolo PUP.

XEROX

Memorándum interno de Metcalfe sobre Ethernet

ETHER!

Xerox Sensitive

MEMO

MAY 22, 1973

TO: ALTO ALOHA DISTRIBUTION

FROM: BOB METCALFE

SUBJECT: ETHER ACQUISITION

HERE IS MORE ROUGH STUFF ON THE ALTO ALOHA NETWORK.

I PROPOSE WE STOP CALLING THIS THING "THE ALTO ALOHA NETWORK". FIRST, BECAUSE IT SHOULD SUPPORT ANY NUMBER OF DIFFERENT KINDS OF STATION -- SAY, NOVA, PDP-11, ..... SECOND, BECAUSE THE ORGANIZATION IS BEGINNING TO LOOK VERY MUCH MORE BEAUTIFUL THAN THE ALOHA RADIO NETWORK -- TO USE CHARLES'S "BEAUTIFUL".

MAYBE: "THE ETHER NETWORK". SUGGESTIONS?

LAZY SUZAN
BULLETIN BOARD
PARLEY
PARLIAMENTARY PROCEDURE

I HOPE TO BE SIMULATING SOON. HELP? INPUTS?

Documento fornecido pelo acervo digital do Museu da Computação

# Estación de trabajo Alto de Xerox (1973)

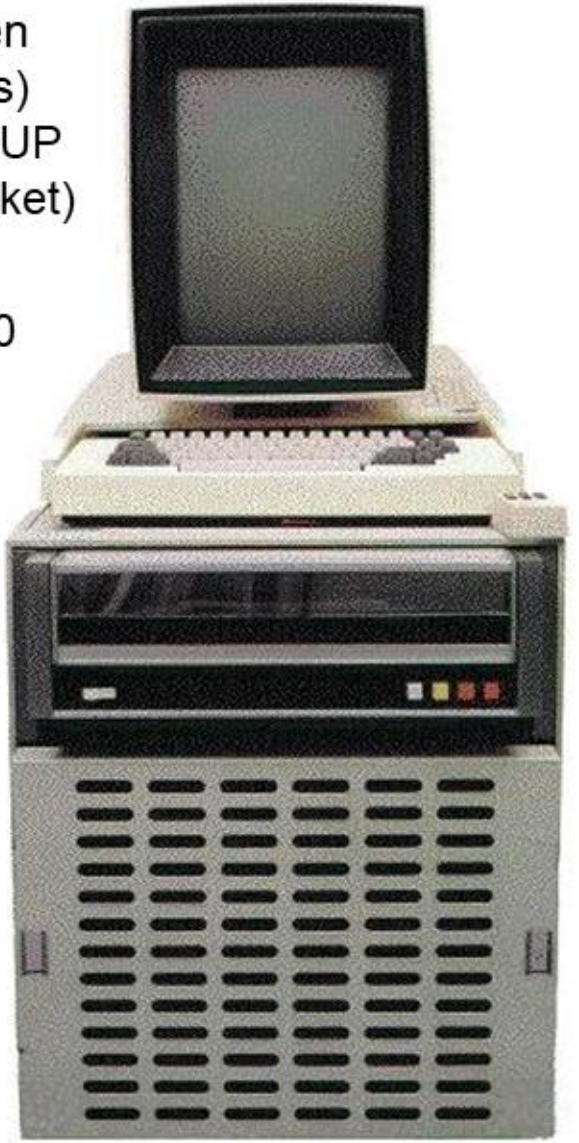
Primer ordenador en Ethernet (2,94 Mb/s)
Utilizaba protocolo PUP (PARC Universal Packet)

Resolución: 800 x 600

CPU: 5,88 MHz (Tres tarjetas, 600 chips)

Memoria: 128 KB
Disco: 2,5 MB

Precio en 1974: 40.000 dólares





David Boggs con una tarjeta Ethernet del Alto

# Lanzamiento comercial de Ethernet: Consorcio DIX

- 1976: Metcalfe y Boggs publican un artículo sobre Ethernet en ACM Communications
- 1976 Xerox crea una nueva división para el lanzamiento comercial de los PCs y de Ethernet, y cambia el nombre a X-Wire, sin éxito.
- 1979 Metcalfe abandona Xerox y crea el consorcio DIX (Digital-Intel-Xerox) para potenciar el uso de Ethernet. A continuación crea 3Com. Empieza movimientos para crear el proyecto 802 en el IEEE
- Sep-1980 DIX publica Ethernet v 1.0, de 92 págs. (el estándar actual tiene 4806 págs.)

Received May 1975; revised December 1975

Communications of the ACM | July 1976 Volume 19 Number 7

Computer Systems | G. Bell, S. Fuller and D. Siewiorek, Editors

# Ethernet: Distributed Packet Switching for Local Computer Networks

Robert M. Metcalfe and David R. Boggs
Xerox Palo Alto Research Center

**Ethernet is a branching broadcast communication system for carrying digital data packets among locally distributed computing stations. The packet transport mechanism provided by Ethernet has been used to build systems which can be viewed as either local computer networks or loosely coupled multiprocessors. An Ethernet's shared communication facility, its Ether, is a passive broadcast medium with no central control. Coordination of access to the Ether for packet broadcasts is distributed among the contending transmitting stations using controlled statistical arbitration. Switching of packets to their destinations on the Ether is distributed**

## 1. Background

One can characterize distributed computing as a spectrum of activities varying in their degree of decentralization, with one extreme being remote computer networking and the other extreme being multiprocessing. Remote computer networking is the loose interconnection of previously isolated, widely separated, and rather large computing systems. Multiprocessing is the construction of previously monolithic and serial computing systems from increasingly numerous and smaller pieces computing in parallel. Near the middle of this spectrum is local networking, the interconnection of computers to gain the resource sharing of computer networking and the parallelism of multiprocessing.

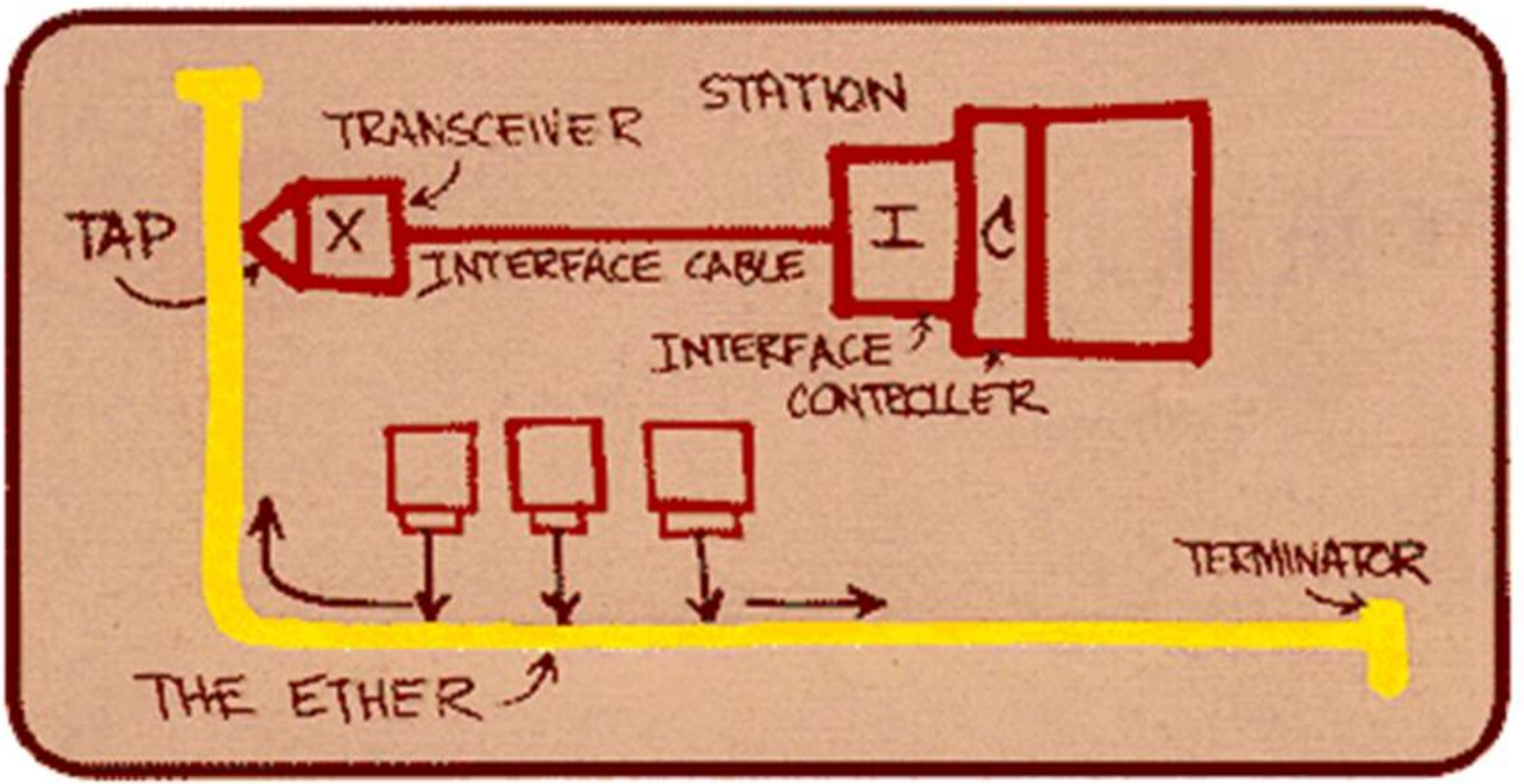
The separation between computers and the associated bit rate of their communication can be used to divide the distributed computing spectrum into broad activities. The product of separation and bit rate, now about 1 gigabit-meter per second (1 Gbmps), is an indication of the limit of current communication technology and can be expected to increase with time:

| Activity | Separation | Bit rate |
|---|---|---|
| Remote networks | > 10 km | < .1 Mbps |
| Local networks | 10–.1 km | .1–10 Mbps |
| Multiprocessors | < .1 km | > 10 Mbps |

### 1.1 Remote Computer Networking

Computer networking evolved from telecommunications *terminal-computer* communication, where the object was to connect remote terminals to a central computing facility. As the need for *computer-computer* interconnection grew, computers themselves were used

# Dibujo de Ethernet hecho por Metcalfe en 1976



Memorándum interno de 1976
sobre la arquitectura de Ethernet
(por entonces 'Xerox-Wire')

**Memorandum**

| | | | |
|---|---|---|---|
| To | Distribution | Date | 21 December 1976 |
| From | Bob Metcalfe | Location | Palo Alto |
| Subject | **Xerox Wire Functional Specification** | Organization | SDD/SD |

XEROX

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Unlike IBM's Systems Network Architecture (SNA), the OIS Protocol is based on distributed many-to-many communication as required by incrementally grown and increasingly interconnected office systems rather than hierarchical mainframe-centered data processing systems. TAKE THAT.

Primer estándar de Ethernet (v 1.0)

# The Ethernet

A Local Area Network

Data Link Layer
and
Physical Layer
Specifications










Digital Equipment Corporation
Maynard, MA

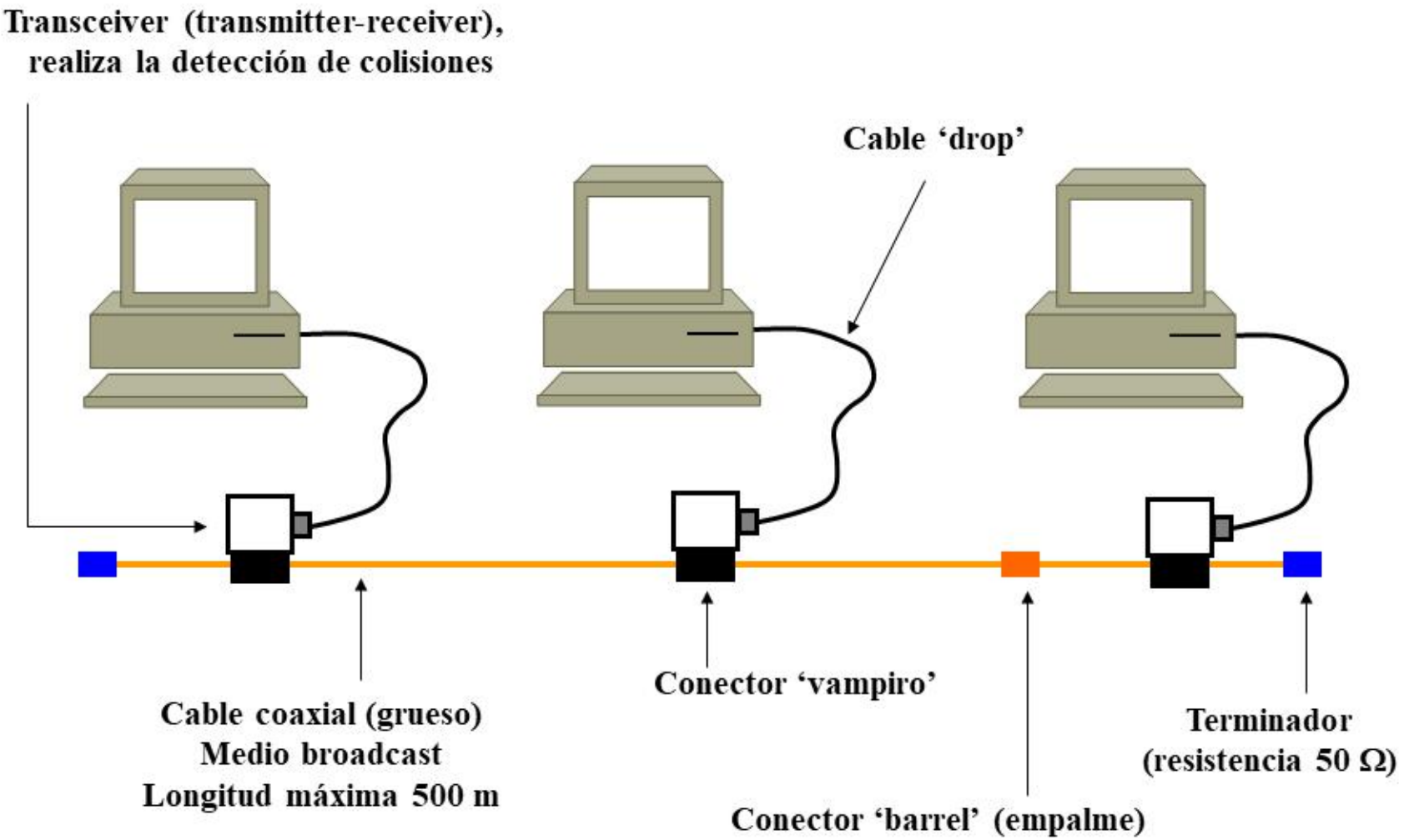
Intel Corporation
Santa Clara, CA

Xerox Corporation
Stamford, CT

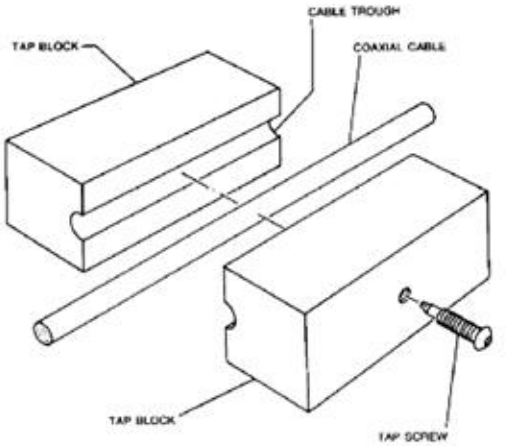
Version 1.0

September 30, 1980

# Ethernet original (10BASE5, 1980-1990)



# Conector 'vampiro' de Ethernet 10BASE5




NOTE: **Tutorial only** and **not** part of specification

**Figure 8-8—Coaxial tap connector configuration concepts**

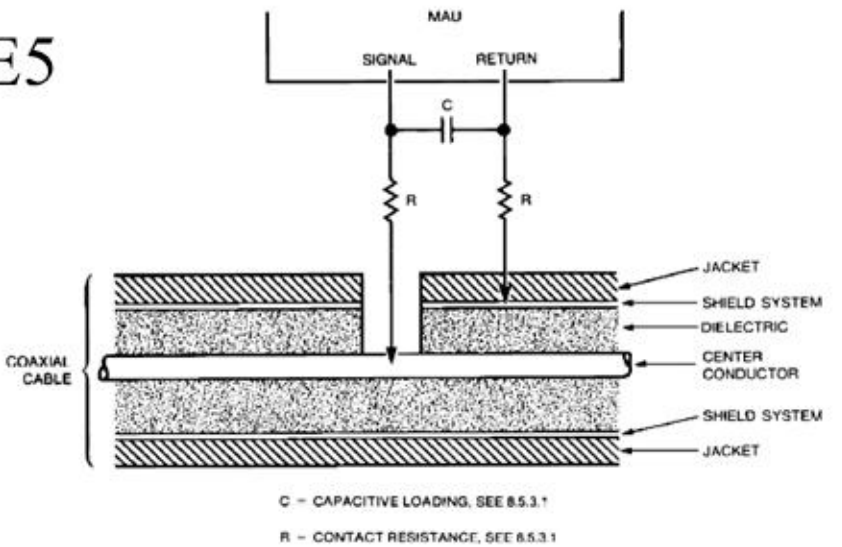
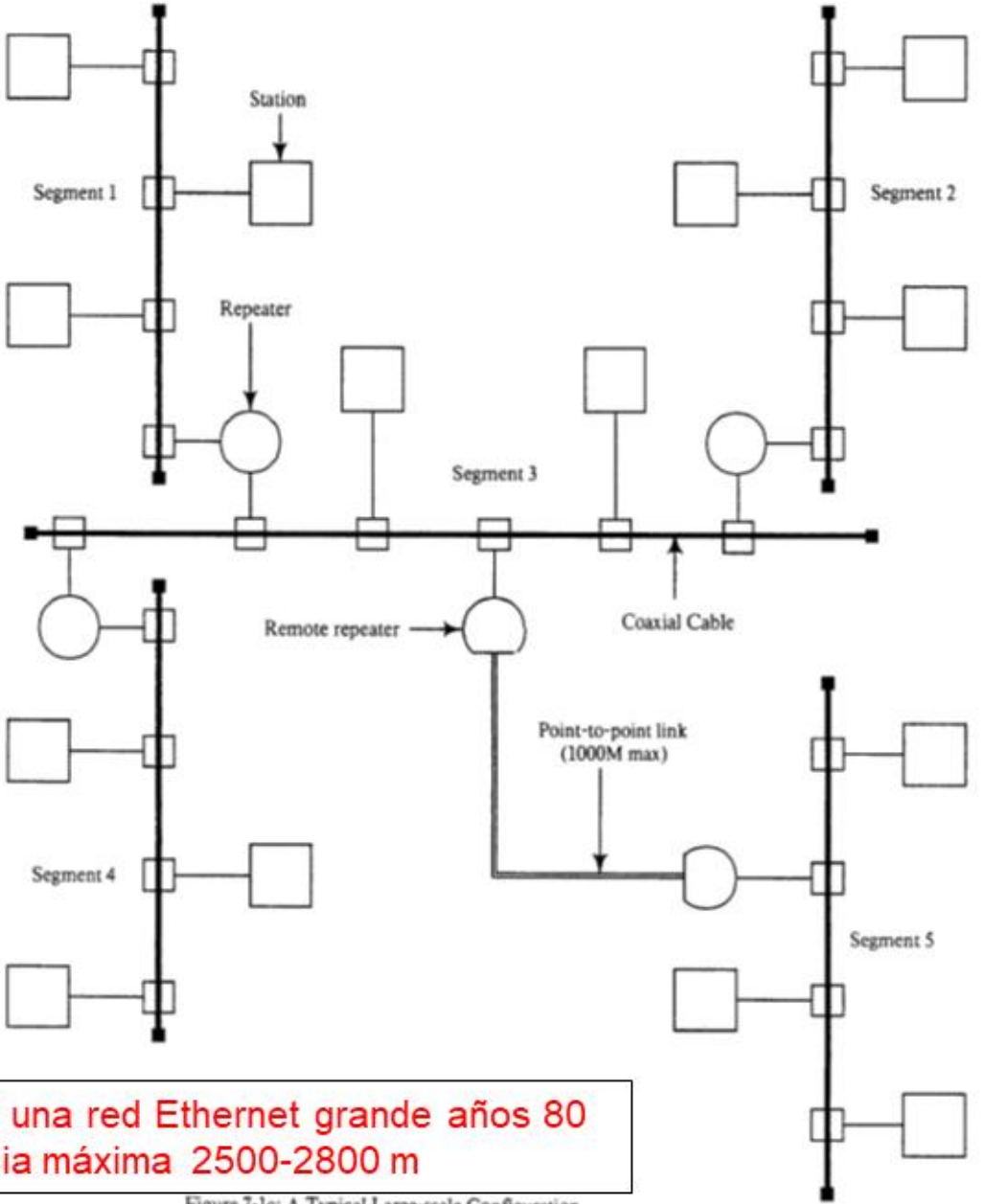



**Figure 8-9—Typical coaxial tap connection circuit**




Configuración de una red Ethernet grande años 80
Distancia máxima 2500-2800 m

Figure 7-1c: A Typical Large-scale Configuration